

Math 54 Discussion Worksheet

Name: _____ ID: _____

Problem 1

Suppose $P_{\mathcal{B} \rightarrow \mathcal{C}}$ is any matrix such that for every vector $v \in V$,

$$[v]_{\mathcal{C}} = P_{\mathcal{B} \rightarrow \mathcal{C}}[v]_{\mathcal{B}} \quad \text{for each } v \in V$$

Set $v = b_1$ in the equation above. Then this shows that $[b_1]_{\mathcal{C}}$ is the first column of $P_{\mathcal{B} \rightarrow \mathcal{C}}$, because _____. Similarly, for $k = 2, \dots, n$, the k -th column of Q is _____, because _____. This shows that the change of basis matrix is unique.

Problem 2

- a) Show that λ is an eigenvalue of A iff it is an eigenvalue of A^T .
- b) Consider an $n \times n$ matrix A such that the rowsums all equal the same number s . Show that s is an eigenvalue of A .
- c) Let λ be an eigenvalue of A . Find a matrix B such that the eigenspace of A associated to λ is the null space of B .

Problem 3

Let A be an $n \times n$ matrix with eigenvalues $\lambda_1, \dots, \lambda_n$ repeated according to multiplicity so that

$$\det(A - \lambda I) = (\lambda - \lambda_1) \cdots (\lambda - \lambda_n).$$

Show that $\det(A)$ is the product of the eigenvalues of A .